

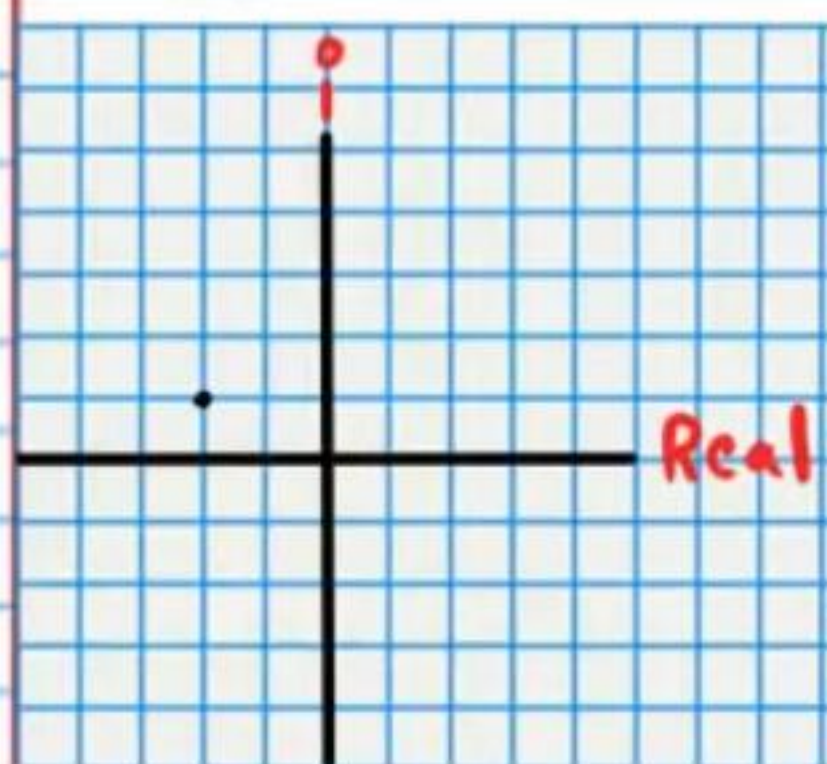
Practice Final Test-Part II (Version 1)

Name
Date
Course

- I) This exam covers the second half of the course (content on Test III & Test IV).
- II) Write your name, the date, and the course title (i.e. Precalculus).
- III) Show all work.
- IV) Draw a rectangle around your final answer for each problem.
- V) A graphing calculator is required.

① Graph the following complex number.

$$-2 + i$$



② For the function below, find an algebraic expression representing the difference quotient (use "h" for "change in x"). It is not necessary to simplify the expression.

$$f(x) = \frac{\tan x}{x}$$

$$DQ = \frac{f(x+h) - f(x)}{h}$$

$$DQ = \frac{\frac{\tan(x+h)}{x+h} - \frac{\tan x}{x}}{h}$$

③ If $f(x) = x^2$, what is the average rate of change of the function per unit x from $x=1$ to $x=3$?

$$M_{ave} = \frac{y_2 - y_1}{x_2 - x_1}$$

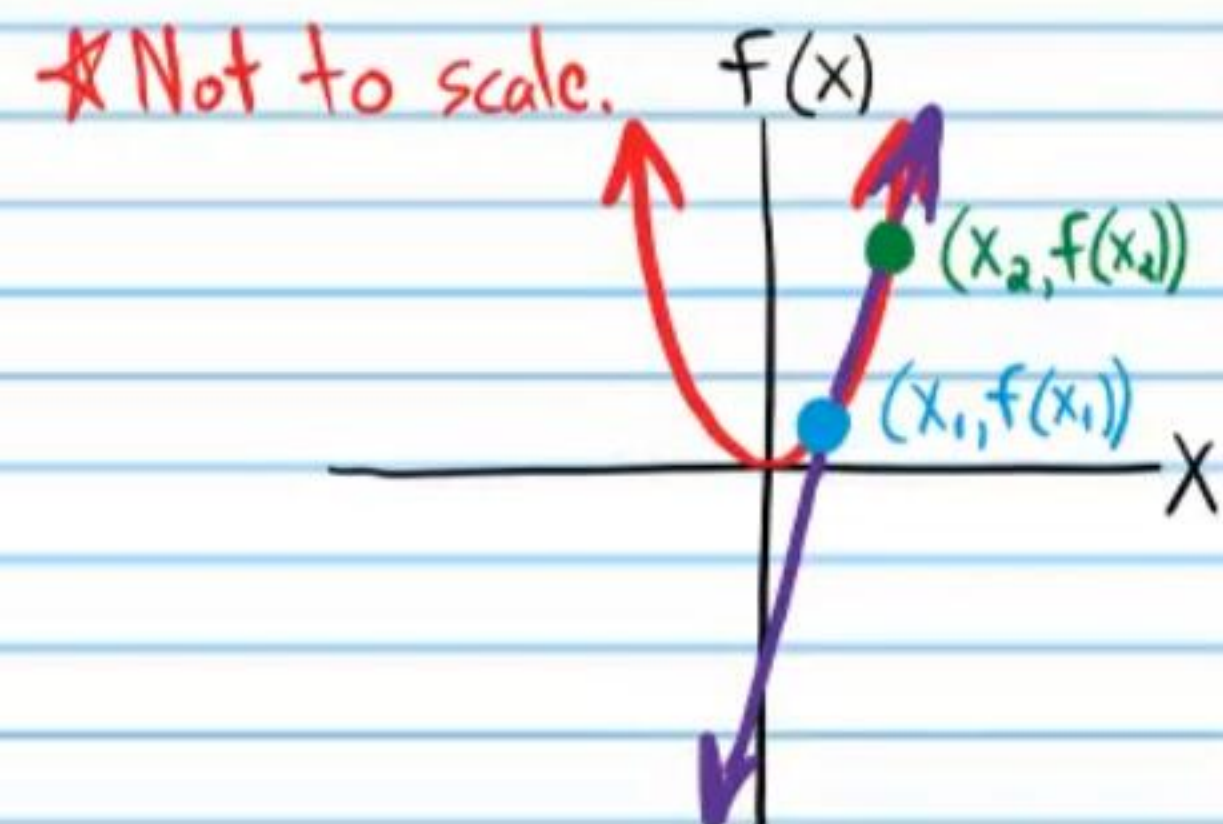
$$= \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$= \frac{f(3) - f(1)}{3 - 1}$$

$$= \frac{3^2 - 1^2}{3 - 1}$$

$$= \frac{9 - 1}{2}$$

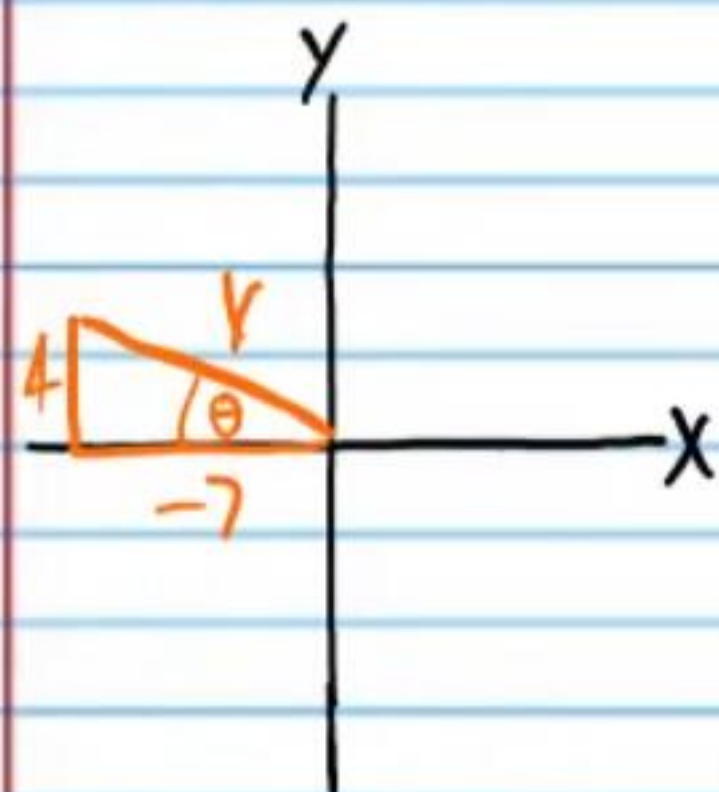
$$= \boxed{4}$$



④ Convert the rectangular coordinates below to polar coordinates.
Use degree units for angle measures. Round to the nearest tenth if necessary.

$(-7, 4)$

Graph



Find r .

$$x^2 + y^2 = r^2$$

$$(-7)^2 + 4^2 = r^2$$

$$49 + 16 = r^2$$

$$65 = r^2$$

$$r = \pm\sqrt{65}$$

$$\boxed{(-\sqrt{65}, -29.7^\circ)}$$

Find θ .

$$\tan\theta = \frac{y}{x}$$

$$\tan\theta = \frac{4}{-7}$$

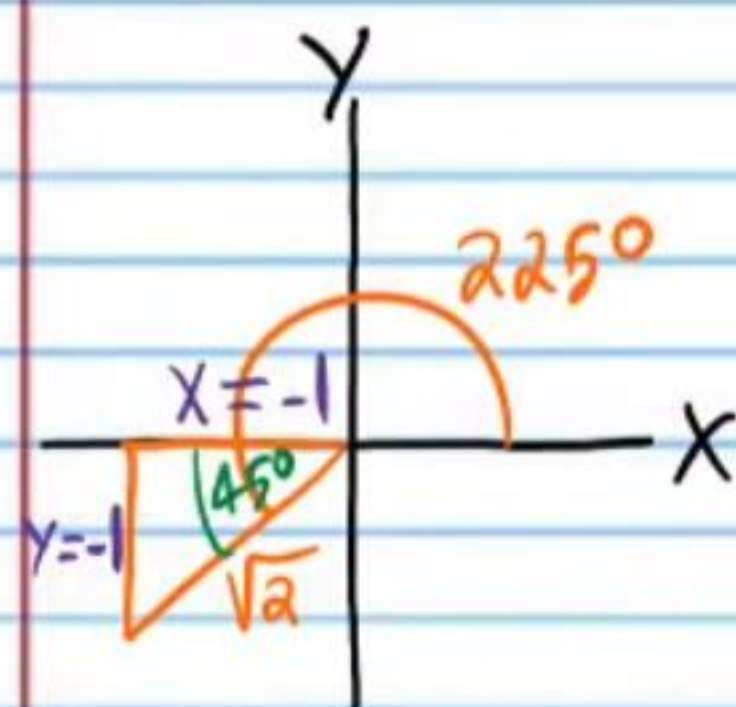
$$\theta = \tan^{-1}\left(-\frac{4}{7}\right)$$

$$\theta \approx -29.7^\circ$$

⑤ Convert the following polar coordinates to rectangular coordinates without a calculator.

$(\sqrt{2}, 225^\circ)$

Graph



Find x . Find y .

$$\boxed{(-1, -1)}$$

⑥ Convert the equation below to polar form.

$$x + y = x^2 + y^2$$

$$r\cos\theta + r\sin\theta = (r\cos\theta)^2 + (r\sin\theta)^2$$

$$r\cos\theta + r\sin\theta = r^2\cos^2\theta + r^2\sin^2\theta$$

$$r(\cos\theta + \sin\theta) = r^2(\cos^2\theta + \sin^2\theta)$$

$$r(\cos\theta + \sin\theta) = r^2(1)$$

$$\boxed{\cos\theta + \sin\theta = r}$$

⑦ If $f(x) = -5x^2 + 11x - 10$ and $f(k) = -5$, find k .

$$f(x) = -5x^2 + 11x - 10$$

$$f(k) = -5k^2 + 11k - 10$$

$$-5 = -5k^2 + 11k - 10$$

$$0 = -5k^2 + 11k - 5$$

$$k = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$k = \frac{-11 \pm \sqrt{11^2 - 4(-5)(-5)}}{2(-5)}$$

$$k = \frac{-11 \pm \sqrt{121 - 100}}{-10}$$

$$k = \frac{-11 \pm \sqrt{21}}{-10}$$

$$\boxed{k = \frac{11 \pm \sqrt{21}}{10}}$$

⑧ Create parametric equations for x and y using t as the parameter.

$$y = \log_2(x+3)$$

$$\boxed{x=t, y=\log_2(t+3)}$$

⑨ Relations are written below in various forms. Write each inverse relation in its corresponding form.

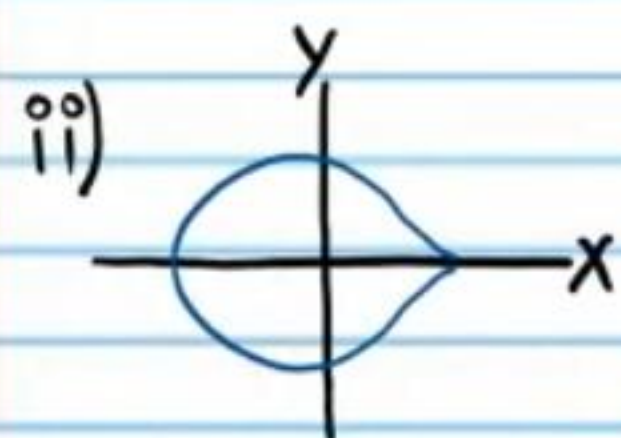
i)

X	14	0	-12	20	-9
Y	-1	8	-9	22	-8

 Write the inverse relation $\rightarrow$

X	-1	8	-9	22	-8
Y	14	0	-12	20	-9

i.e. Reflection across $y=x$



Write the inverse relation
i.e. Reflection across $y=x$



iii) $\csc^2 x - 1 = y$

Write the inverse relation
i.e. Reflection across $y=x$

$$\boxed{\csc^2 y - 1 = x}$$

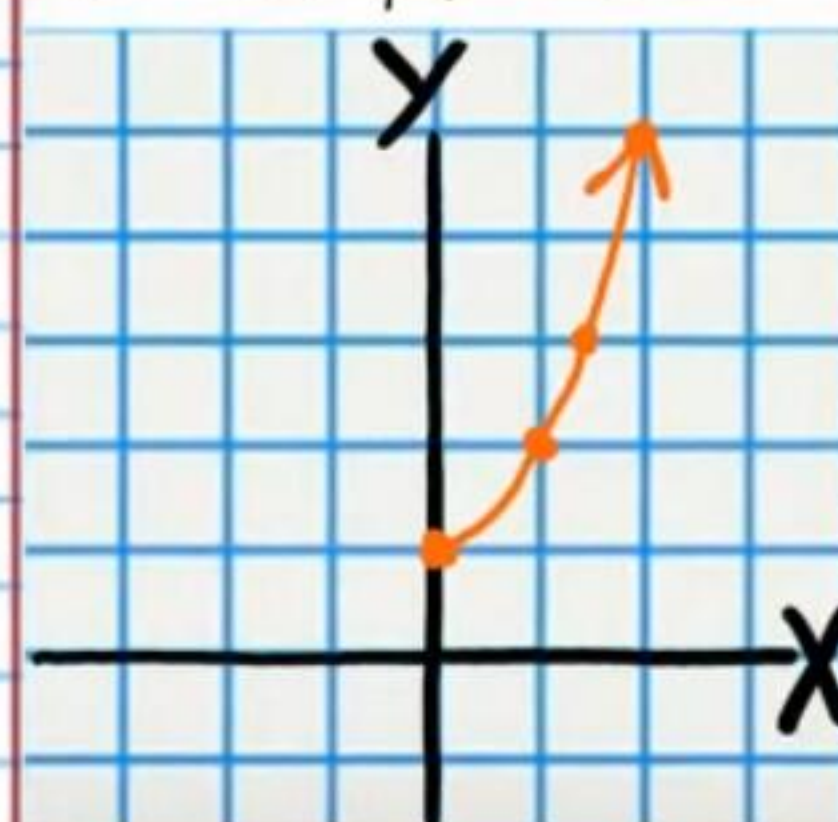
iv) $|x| + |y-1| = y$

Write the inverse relation
i.e. Reflection across $y=x$

$$\boxed{|y| + |x-1| = x}$$

⑩ Graph the following equation. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

$$x = \sqrt{t}, y = t+1$$



Point #1

$$t=0$$

$$x = \sqrt{0}, y = 0+1$$

$$x = 0, y = 1$$

$$\boxed{(0,1)}$$

Point #2

$$t=1$$

$$x = \sqrt{1}, y = 1+1$$

$$x = 1, y = 2$$

$$\boxed{(1,2)}$$

Point #3

$$t=4$$

$$x = \sqrt{4}, y = 4+1$$

$$x = 2, y = 5$$

$$\boxed{(2,5)}$$

Point #4

$$t=2$$

$$x = \sqrt{2}, y = 2+1$$

$$\boxed{(\sqrt{2}, 3)}$$

Eliminate the parameter

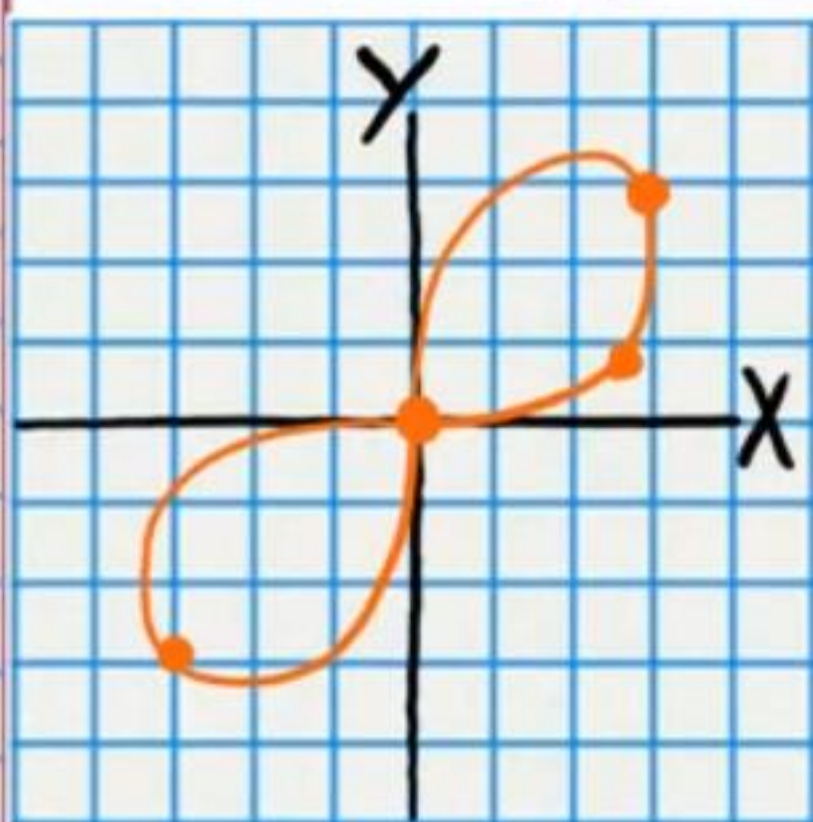
$$x = \sqrt{t}, y = t+1$$

$$x^2 = t$$

$$\boxed{y = x^2 + 1}$$

11) Graph the following equation. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$$r = \sqrt{16 \sin(2\theta)}$$



* Hint: We are not allowed to have a negative number under a square root symbol, so we must choose θ values between 0° & 90° or between 180° & 270° .

1st Point

$$\theta = 45^\circ$$

$$r = \sqrt{16 \sin(2(45^\circ))}$$

$$r = \sqrt{16 \sin 90^\circ}$$

$$r = \sqrt{16(1)}$$

$$r = 4$$

$$(4, 45^\circ)$$

2nd Point

$$\theta = 0^\circ$$

$$r = \sqrt{16 \sin(2(0^\circ))}$$

$$r = \sqrt{16(0)}$$

$$r = 0$$

$$(0, 0^\circ)$$

4th Point

$$\theta = 225^\circ$$

$$r = \sqrt{16 \sin(2(225^\circ))}$$

$$r = \sqrt{16 \sin 450^\circ}$$

$$r = \sqrt{16(1)}$$

$$r = 4$$

$$(4, 225^\circ)$$

3rd Point

$$\theta = 15^\circ$$

$$r = \sqrt{16 \sin(2(15^\circ))}$$

$$r = \sqrt{16 \sin 30^\circ}$$

$$r = \sqrt{16\left(\frac{1}{2}\right)}$$

$$r = 2\sqrt{2}$$

$$(2\sqrt{2}, 15^\circ)$$

12) Perform the following reflections.

i)	X	-6	0	2	17	20
	Y	-5	4	-1	10	40

Reflect across the x-axis

X	-6	0	2	17	20
Y	5	-4	1	-10	-40

ii) $y^2 = x - 2x^2$

Reflect across the y-axis

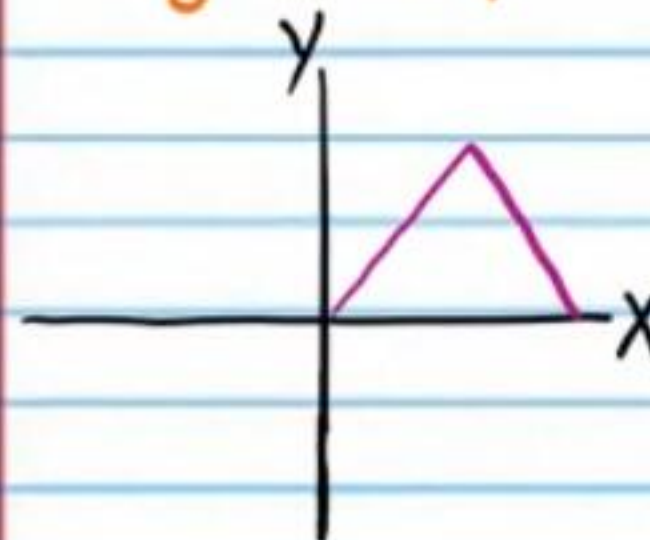
$$y^2 = -x - 2(-x)^2$$

$$y^2 = -x - 2x^2$$

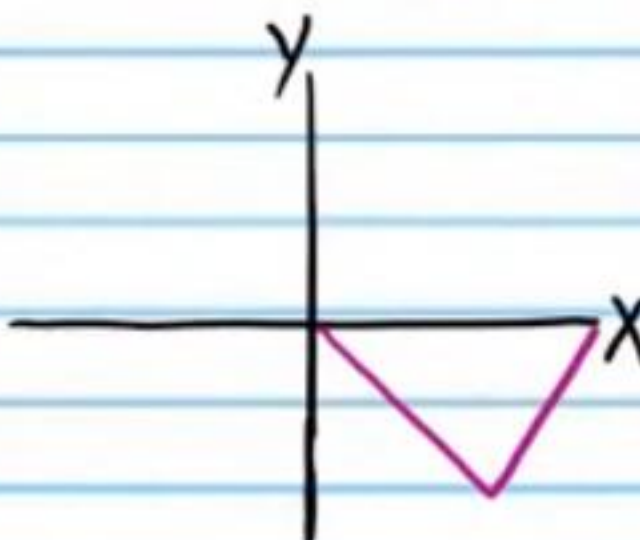
iii)

Reflect across the origin

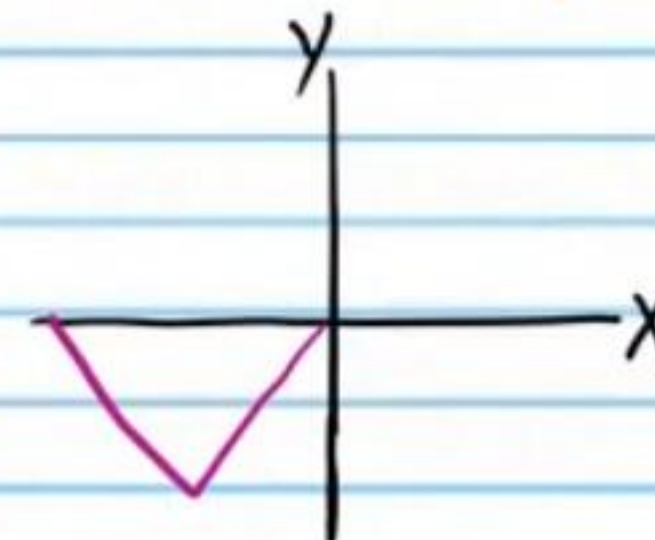
Original Graph



Step 1
Reflect across the x-axis



Step 2
Reflect across the y-axis

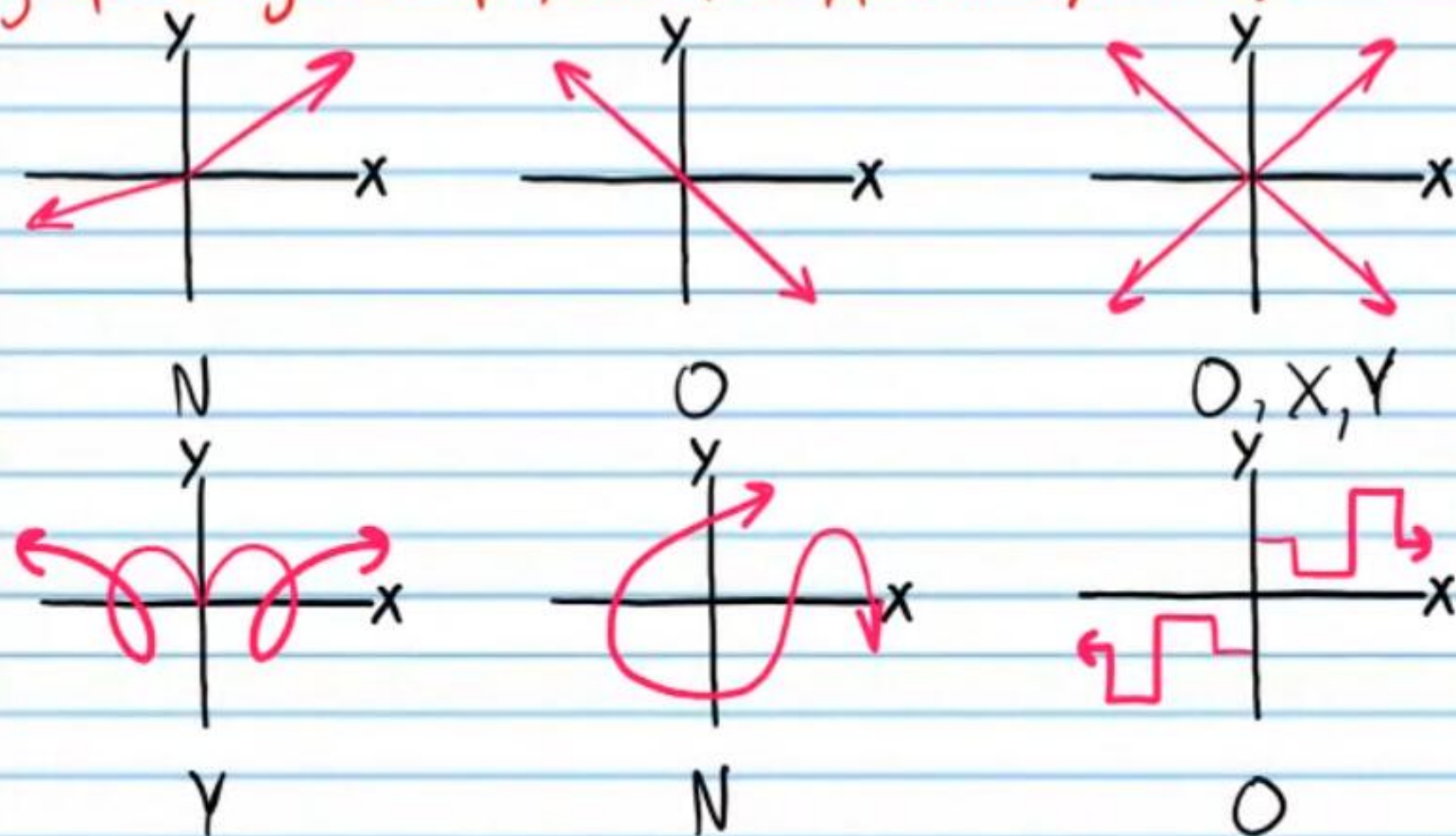


iv) $\sin x^2 = y + x$

Reflect across the x-axis

$$\sin x^2 = -y + x$$

13) Relations are written below in the form of graphs. If a relation appears to be symmetric about the x-axis, write "x". If a relation appears to be symmetric about the y-axis, write "y". If a relation appears to be symmetric about the origin, write "O". If the relation shows no apparent symmetry, write "N". *Some graphs might display multiple types of symmetry.*



ii) The table below has points collected from a graph that is symmetric about the x-axis. Fill the empty spaces with other points from the corresponding graph.

X	10	13	11	18	-9	10	13	11	18	-9
Y	9	8	-2	-18	14	-9	-8	2	18	-14

Use the following equation to answer the questions below.

$$y^2 + x = \cos y$$

iii) Is the graph of the equation symmetric about the y-axis? *Demonstrate algebraically.*

$$y^2 - x = \cos y$$

No

iv) Is the graph of the equation symmetric about the origin? *Demonstrate algebraically.*

$$(-y)^2 - x = \cos(-y)$$

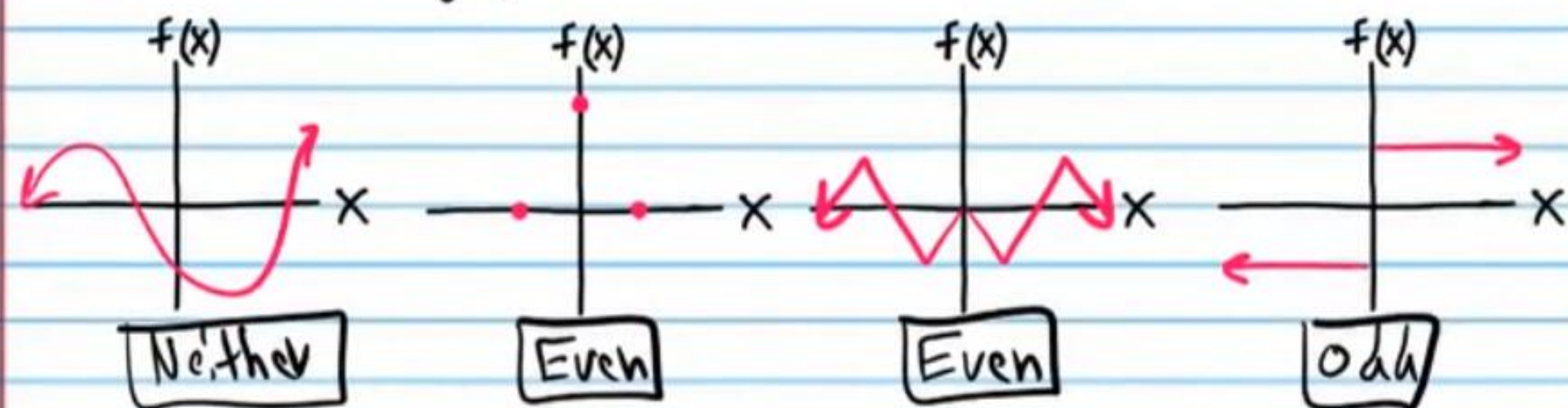
$$y^2 - x = \cos y$$

No

14) The table below has points collected from a graph of an odd function. Fill the empty spaces with other points from the graph.

X	1	2	8	4	11	-1	-2	-8	-4	-11
f(x)	2	3	7	3	12	-2	-3	-7	-3	-12

ii) Determine if the graphs below are even functions, odd functions, or neither.



iii) If "f" is an even function, and $f(-2)=11$, find $f(2)$.

$$(-2, 11) \rightarrow (2, 11) \quad f(2) = \boxed{11}$$

iv) If "f" is an odd function, and $f(9)=346$, find $f(-9)$.

$$(9, 346) \rightarrow (-9, -346) \quad f(-9) = \boxed{-346}$$

15) Determine if the function below is even, odd, or neither.

$$f(x) = \frac{x^2}{\sec x}$$

$$f(-x) = \frac{(-x)^2}{\sec(-x)}$$

$$f(-x) = \frac{x^2}{\sec x}$$

$$f(-x) = f(x)$$

Even

$$f(-x) = f(x) \rightarrow \text{even}$$

$$f(-x) = -f(x) \rightarrow \text{odd}$$

16) If $f(x) = -2x^2 - 4x - 1$ & $h(x) = -2x^2 + x + 1$, for what value(s) of x does $f(x) = h(x)$?

$$f(x) = h(x)$$

$$-2x^2 - 4x - 1 = -2x^2 + x + 1$$

$$-4x - 1 = x + 1$$

$$x = -\frac{2}{5}$$

17) The population of a colony of bacteria after "t" hours can be predicted with the equation below.

$$P(t) = \frac{99,000}{1 + 36e^{-0.13t}}$$

i) What is the carrying capacity of the population?

99,000 bacteria

ii) What is the initial population?

$$P(0) = \frac{99,000}{1 + 36e^{-0.13(0)}} = \frac{99,000}{1 + 36e^0} = \frac{99,000}{1 + 36(1)}$$

$$= \frac{99,000}{37}$$

$$\approx \boxed{2,676 \text{ bacteria}}$$

iii) What will the population be after 5 hours?

$$P(5) = \frac{99,000}{1 + 36e^{-0.13(5)}} = \boxed{5,002 \text{ bacteria}}$$

iv) Use your calculator's base "e" logarithm button (LN) and Logarithm Property IX to find the value of the logarithm below. Round to the nearest thousandth.

$$\log_4 11 = \frac{\log_e 11}{\log_e 4} = \frac{\ln 11}{\ln 4} \approx \boxed{1.73}$$

(18) i) Relations are written below in table form. Some of the tables could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. For each table, write "1-to-1 function," "function," or "not a function."

X	36	-25	40	41	22
Y	42	89	46	46	26

Function

X	0	13	8	7	0
Y	1	12	10	9	2

Not a Function

X	-30	-16	-15	-20	-10
Y	-20	-8	-10	-9	-1

1-to-1 Function

X	32	18	20	15	10
Y	45	41	44	41	41

Function

ii) In the following problem, "f" and "f⁻¹" are inverse functions.

If $f(-1) = 9$, find $f^{-1}(9)$.

$(-1, 9)$ $(9, -1)$ $f^{-1}(9) = \boxed{-1}$

(19) In 14 years, a radioactive substance decreases from 600 grams to 85 grams. How much of this same radioactive material would be left if we started with a 316 gram sample and monitored it for 10 years?

I) Find k.

$$A(t) = A_0 e^{kt}$$

$$85 = 600 e^{k(14)}$$

$$85 = 600 e^{14k}$$

$$\frac{85}{600} = \frac{600 e^{14k}}{600}$$

$$\frac{85}{600} = e^{14k}$$

$$14k = \log_e \left(\frac{85}{600} \right)$$

$$14k = \ln \left(\frac{85}{600} \right)$$

$$k = \frac{\ln \left(\frac{85}{600} \right)}{14}$$

$$k \approx -0.13959...$$

II) $A(t) = A_0 e^{kt}$

$$A(10) = 316 e^{(-0.13959...)(10)}$$

$$A(10) \approx \boxed{78.24 \text{ grams}}$$

20) Find the equation of the inverse function. Restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

$$f(x) = (x-1)^2, \quad \boxed{x \geq 1}$$

$$\downarrow$$

$$\text{I) } y = (x-1)^2$$

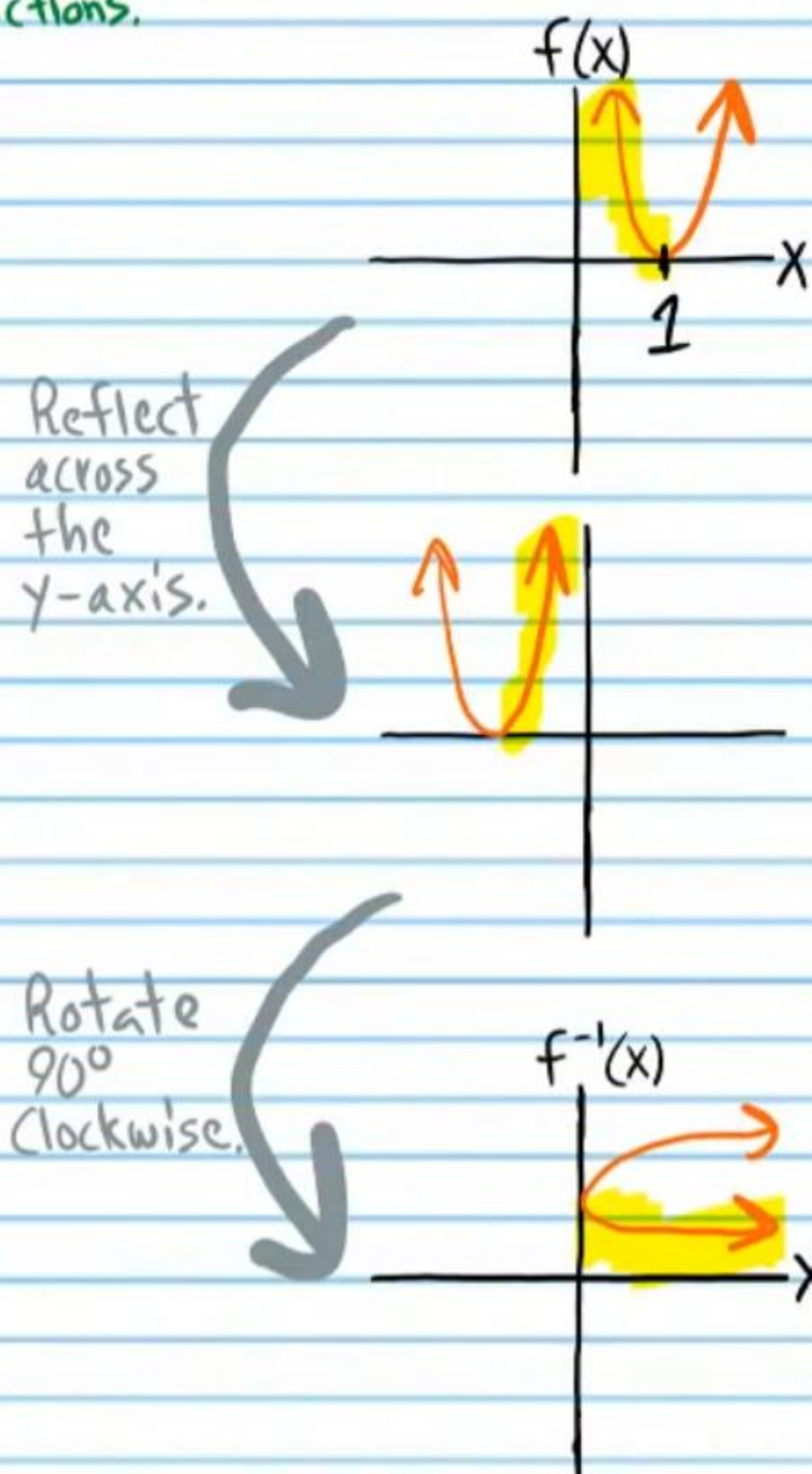
$$\text{II) } x = (y-1)^2$$

$$\text{III) } \pm\sqrt{x} = y-1$$

$$y = \pm\sqrt{x} + 1$$

$$\downarrow$$

$$\text{IV) } \boxed{f^{-1}(x) = \sqrt{x} + 1}$$



For problem #'s 21-22, find an equation of the inverse function.

$$\textcircled{21} f(x) = 4(10^x) - 3$$

$$\downarrow$$

$$\text{I) } y = 4(10^x) - 3$$

$$\text{II) } x = 4(10^y) - 3$$

$$\text{III) } x+3 = 4(10^y)$$

$$\frac{x+3}{4} = 10^y$$

$$y = \log_{10}\left(\frac{x+3}{4}\right)$$

$$y = \log\left(\frac{x+3}{4}\right)$$

$$\downarrow$$

$$\text{IV) } \boxed{f^{-1}(x) = \log\left(\frac{x+3}{4}\right)}$$

$$\textcircled{22} f(x) = \frac{\ln x}{26} + 4$$

$$\downarrow$$

$$\text{I) } y = \frac{\log_e x}{26} + 4$$

$$\text{II) } x = \frac{\log_e y}{26} + 4$$

$$\text{III) } x-4 = \frac{\log_e y}{26}$$

$$26x - 104 = \log_e y$$

$$y = e^{26x - 104}$$

$$\downarrow$$

$$\text{IV) } \boxed{f^{-1}(x) = e^{26x - 104}}$$

②③ Show algebraically that "f" and "g" are inverse functions.

$$f(x) = 2x - 3 \quad \& \quad g(x) = \frac{x+3}{2}$$

$$f(g(x)) = f\left(\frac{x+3}{2}\right) = 2\left(\frac{x+3}{2}\right) - 3 = x + 3 - 3 = x$$

✓

$$g(f(x)) = g(2x - 3) = \frac{(2x - 3) + 3}{2} = \frac{2x}{2} = x$$

✓

Yes

②④ Find the intensity of sound required to produce the following decibels.

60 dB

$$L(x) = 10 \log_{10}(10^{12}x)$$

$$L(x) = 10 \log_{10}(10^{12}x)$$

$$60 = 10 \log_{10}(10^{12}x)$$

$$\frac{60}{10} = \frac{10 \log_{10}(10^{12}x)}{10}$$

$$6 = \log_{10}(10^{12}x)$$

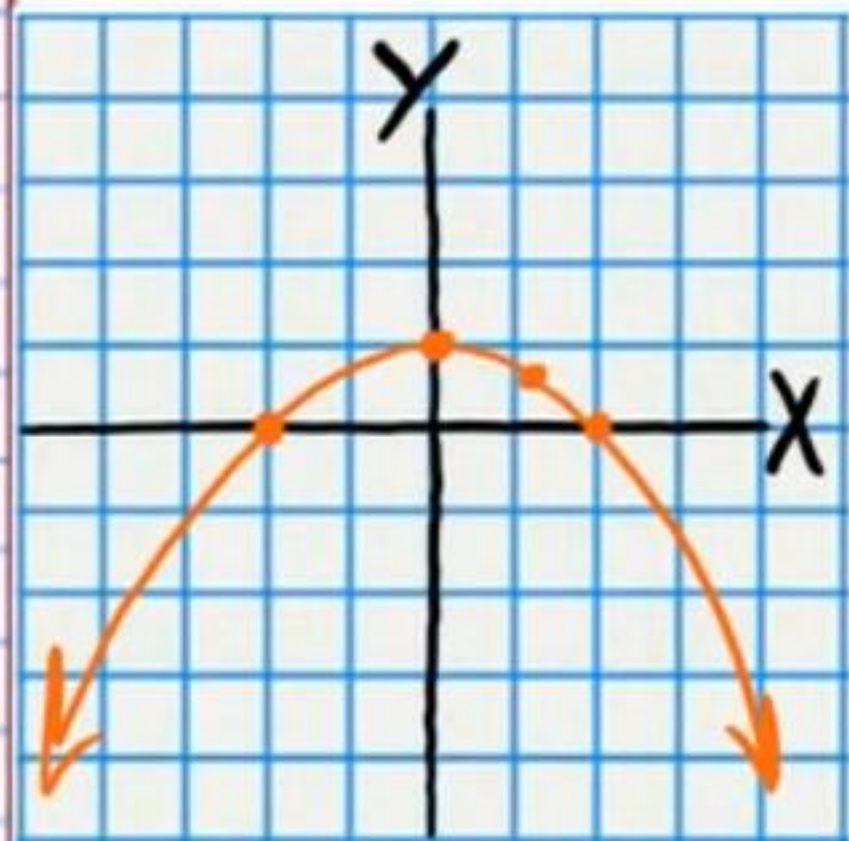
$$10^{12}x = 10^6$$

$$\frac{10^{12}x}{10^{12}} = \frac{10^6}{10^{12}}$$

$$x = 10^{-6} \text{ W/m}^2$$

25) Graph the following equation using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$$r = \frac{2}{1 + \sin \theta}$$



1st Point

$$\theta = 0$$

$$r = \frac{2}{1 + \sin 0}$$

$$r = 2$$

$$(2, 0)$$

2nd Point

$$\theta = \frac{\pi}{2}$$

$$r = \frac{2}{1 + \sin \frac{\pi}{2}}$$

$$r = \frac{2}{1 + 1}$$

$$r = 1$$

$$(1, \frac{\pi}{2})$$

3rd Point

$$\theta = \pi$$

$$r = \frac{2}{1 + \sin \pi}$$

$$r = \frac{2}{1 + 0}$$

$$r = 2$$

$$(2, \pi)$$

4th Point

$$\theta = \frac{\pi}{6}$$

$$r = \frac{2}{1 + \sin \frac{\pi}{6}}$$

$$r = \frac{2}{1 + \frac{1}{2}}$$

$$r = \frac{2}{\frac{3}{2}}$$

$$r = 2 \div \frac{3}{2}$$

$$r = \frac{4}{3}$$

$$(\frac{4}{3}, \frac{\pi}{6})$$